Learning Management System Using LLaMa

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CERTIFICATE OF APPROVAL

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a. The work contained in this project has been done by us under the guidance of our supervisor(s).
b. The work has not been submitted to any other Institute for any degree or diploma.
c. We have followed the guidelines provided by the Institute in preparing the project report.
d. We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
e. Whenever we have used materials (data, theoretical analysis, figures, and text) from other sources, we have given due credit to them by citing them in the text of the report and giving their details in the references. Further, we have taken permission from the copyright owners of the sources, whenever necessary.

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ABSTRACT

The proposed Learning Management System (LMS) introduces an innovative solution for intelligent search and task optimization by integrating LLaMA, a cutting-edge generative AI model. This system enhances data retrieval and resource management through a smart decision agent that intelligently routes user queries, ensuring high relevance and precision. It further incorporates automated topic decomposition, enabling the generation of personalized learning roadmaps tailored to each student's knowledge level and learning pace. To support diverse learning styles, the LMS delivers curated video recommendations and continuously refines its suggestions through an adaptive user feedback loop. This dynamic and user-centric approach not only improves engagement but also ensures learners receive contextually appropriate guidance at each stage. Importantly, all components operate within an on-premises environment, safeguarding user data and maintaining strict privacy standards. By offering structured, relevant, and secure learning experiences, this LMS empowers students to navigate complex educational content more effectively and achieve better academic outcomes.

KEYWORDS: Learning Management System (LMS), LLaMa, SearXNG, Ollama, Youtube Integration, Artificial Intelligence (AI), Personalized Learning, Adaptive Learning, Generative AI, Video Recommendation, Decision Agent, User Feedback Loop, Application Programming Interface (API), React, Express.js, Node.js, Large Language Models (LLMs)

1 INTRODUCTION

1.1 Overview

This paper explores the pivotal role of Artificial Intelligence (AI) in transforming the educational landscape, focusing on the significant opportunities it presents to personalize, optimize, and enrich learning experiences. The conventional education system, often bound by rigid structures and a one-size-fits-all approach, is steadily being replaced by adaptive learning models capable of catering to the diverse needs of learners. These modern learning paradigms are fueled by AI, offering scalable, responsive, and learner-centric experiences. At the forefront of this transformation is our proposed Learning Management System (LMS), an AI-powered platform that embodies these advanced educational principles.

The system integrates LLaMA, a powerful generative language model, to offer context-aware educational support to learners. Acting as an intelligent tutor, LLaMA assists users by interpreting their queries, tracking their learning progress, and guiding them through personalized pathways. This integration empowers learners to explore subjects more deeply, seek clarification instantly, and follow learning paths tailored to their unique strengths and weaknesses. It also supports continuous assessment by monitoring user interactions and dynamically adjusting the learning plan, allowing for a more fluid and effective educational journey.

In addition to LLaMA, the LMS includes SearxNG, a privacy-focused metasearch engine that enhances the learning experience by providing access to high-quality, diverse educational resources. SearxNG aggregates content from a wide range of repositories without compromising user privacy. This is particularly important in educational environments where the confidentiality of student data is a growing concern. By functioning as an on-premise solution, the LMS ensures that all user interactions and data remain securely within the institution’s infrastructure, mitigating risks associated with third-party data exposure.

A major advancement introduced in this LMS is the integration of prompt-based course generation using the YouTube API. This feature allows users to enter any custom educational prompt, such as a topic or concept they wish to learn. In response, the system

automatically queries the YouTube API to generate a curated playlist or course outline consisting of high-quality, relevant videos. This dynamic course generation model dramatically enhances flexibility, enabling learners to explore virtually any subject on demand. It also addresses a core limitation of many LMS platforms: reliance on static, pre-uploaded content. With this innovation, content remains current, diverse, and tailored to learners' evolving interests.

To further enhance the system's adaptiveness, the LMS now includes a user feedback collection mechanism after each course suggestion. Once a course is generated based on a user prompt, the system asks for structured feedback — for example, ratings, usefulness, clarity, and relevance of the suggested videos. This feedback not only helps in refining the content recommendations but also contributes to improving the AI model's accuracy over time. As more data is collected, the system evolves, delivering increasingly precise and valuable educational experiences. This loop of user input and AI refinement ensures a continuously improving and learner-sensitive environment.

On the technical front, the LMS is built using the MERN stack — MongoDB, Express.js, React, and Node.js — which offers a robust and scalable infrastructure capable of handling high concurrency and extensive datasets. MongoDB's flexibility in schema design allows for the storage of complex user profiles, learning progress data, and personalized course histories. The Express.js API handles communication between the front end and the server, ensuring smooth request handling and efficient data management. React is used to build an intuitive, responsive, and interactive front-end interface that adapts seamlessly across devices. Node.js powers the backend with event-driven, asynchronous processing for real-time performance.

Security and data privacy remain foundational pillars of this LMS. All user data is encrypted in transit and at rest, and strict access controls ensure that only authorized personnel can access sensitive information. The system is designed to function within institutional networks, maintaining complete data ownership by the deploying organization.

From an educational standpoint, the LMS supports diverse learning styles through a variety of features — project-based tasks, forum-based peer discussions, interactive dashboards, and progress tracking tools. It fosters active participation, critical thinking, and collaboration, moving away from passive, lecture-based instruction. By offering

prompt-based learning, real-time feedback loops, and content sourced from global platforms like YouTube, the LMS accommodates modern learners who value autonomy, instant access, and multimedia-driven instruction.

In conclusion, the integration of advanced AI components such as LLaMA and SearxNG, combined with real-time YouTube-based course generation and learner feedback mechanisms, establishes this LMS as a next-generation solution for education. It not only addresses the limitations of traditional learning environments but also aligns with global trends toward personalized, accessible, and scalable education. The LMS illustrates how AI can be thoughtfully applied to meet the diverse, ever-changing needs of students and educators, paving the way for more intelligent, engaging, and inclusive learning systems.

1.2 Problem statement

Another essential point to state is that LMS have been the only models for online learning for many years, while they can hardly be described as smart learning platforms. These platforms employ a one size fits all model similar to a normal television puts forward where every piece of content is nurtured in one particular way and at one particular speeds there does not seem to be a realization of the variability of learning capabilities in addition to the demographic distribution of the student base. Because of this, these systems merely get at the students in a way that is half way to capturing their attention which leads to loading off and poor performance. Moreover, most of the initial learning management systems are known to support pre-defined learning paths, thus unlike to Blackboard, learning paths, which are rather rigid and do not take into account the students' achievements or their difficulties. Nonetheless content update is typically an affair not very efficient as it is normally done by hand hence demanding massive resources and the system cannot always provide the latest revolutionary content in education or even good content that can suffice the demands of a certain category of learners. This greatly limits the flexibility that instructors have in making the learners undergo the learning experiences they are supposed to go through. Furthermore, most of the common Traditional LMS have some of the following problems: Traditional LMS on the other hand are not capable to provide real time, contextual support to students. The problem arises when students do not understand some of the ideas given in the lectures or when they require a quick explanation of some parts; the students have to

endure the content themselves or wait for the teacher who might react only in several hours or even days. This is so because, due to their lack of adaptive learning mechanisms as found out, it means knowledge delivery will not be prorate in accordance of the disposition given by the students and this exacerbates these problems. Considering the present state of conventional LMS, there is the need for change in the delivery of learning systems. Unfortunately, a traditional, course-centric LMS does not meet these needs and provides little motivation for learners or retention of the knowledge they receive. In specific, future systems using such models of AI can make applications of personalised learning more dynamic and relevant to the 4 needs of students and Faculties. These technologies can offer the adaptability, immediate support and personalized material that is needed to have a highly positive impact on the learning process. Artificial Intelligence (AI) has revolutionized various sectors, and education is no exception. The integration of AI into educational frameworks is reshaping the way knowledge is imparted, accessed, and consumed. AI enables systems to cater to individual learning needs, moving beyond the rigid structures of traditional methods. This transformation is especially visible in Learning Management Systems (LMS), which are becoming increasingly sophisticated with AI technologies. An LMS powered by AI is not merely a repository of educational content but a dynamic platform that adapts to the learner's preferences, pace, and requirements. By analysing patterns in a learner's interaction, AI offers insights into areas of improvement, providing recommendations and resources tailored to their needs. Such advancements underline the pivotal role AI plays in enhancing the educational experience and addressing the diverse challenges of modern learning. The innovative LMS discussed here builds on this foundation, integrating two advanced AI technologies: LLaMA and SearxNG. LLaMA, an acronym for Large Language Model Meta AI, is a sophisticated language model designed to understand and generate human-like text. This capability enables LLaMA to act as an intelligent tutor, capable of explaining complex concepts, answering questions, and engaging learners in meaningful conversations. Complementing this is SearxNG, a privacy-focused metasearch engine that aggregates information from various sources while maintaining user anonymity. By combining the contextual intelligence of LLaMA with the extensive resource retrieval capabilities of SearxNG, this LMS provides a personalized and resource-rich learning environment. These technologies work in synergy to redefine the scope of learning management, making it more interactive, secure, and effective.

1.2.1 Limitations of Current Learning Management System

Despite the widespread adoption of Learning Management Systems (LMS) such as ChatGPT-powered platforms, YouTube educational channels, and Gemini, several inherent limitations adversely impact the overall quality and effectiveness of the learning experience.

One of the primary issues is the generalized, one-size-fits-all approach commonly employed by these platforms. They often fail to account for important individual learner traits, such as the pace of learning, preferred learning styles, and engagement preferences. This lack of personalized consideration can lead to learner disengagement, as users struggle to find courses or content that resonate with their specific needs. For instance, while platforms like YouTube offer vast amounts of educational content, much of it is not organized or aligned with structured lesson plans, resulting in disconnected or fragmented learning paths.

Furthermore, many current LMS platforms tend to reduce direct interaction between students and educators or specialists. This reduced functional interaction limits the opportunity for immediate clarification, personalized guidance, and meaningful feedback—elements essential for fostering deep understanding and competence development. Although advanced tools like ChatGPT can provide information and answer queries, they fall short in delivering real-time, actionable feedback or monitoring ongoing learning processes, which are critical for adapting instruction and improving learner outcomes.

Privacy concerns present another significant challenge. Most widely-used LMS platforms rely on cloud-based storage for user data, which makes sensitive personal and institutional information vulnerable to cyberattacks. Users often have limited control over their personal data, raising issues around consent and data misuse. Additionally, the sheer volume of information transmitted through these platforms can overwhelm learners, who may then selectively engage only with familiar or superficially relevant content, thereby reducing the overall coherence and effectiveness of the learning experience.

A further limitation relates to the modality and freshness of educational resources. Many traditional LMS platforms depend heavily on static, pre-uploaded materials curated by

educators, which can become outdated quickly in fast-evolving academic fields. While some platforms provide links to supplementary resources, these are often limited in scope and do not adequately leverage the vast repository of academic journals, articles, videos, and interactive materials available online. This restriction confines learners to a narrow subset of resources, limiting their ability to explore comprehensive, up-to-date, and diversified content that enhances learning depth and breadth.

Additionally, most existing LMS platforms lack robust mechanisms to ensure data security and privacy. Cloud-based systems, in particular, are vulnerable to hacking, which can result in the compromise of student, faculty, and institutional information. The consequences of such breaches are severe, ranging from legal repercussions to erosion of trust in educational institutions. Many LMS providers also fail to implement stringent security protocols to protect user data during searches or communications, further jeopardizing confidentiality. This is especially critical in educational environments where trust and privacy are paramount for effective communication and content delivery.

Lastly, the absence of privacy-aware tool integration severely restricts the ability of current LMS platforms to create a secure, confidential learning atmosphere. Without incorporating technologies that prioritize user privacy, these platforms struggle to offer learners and educators a protected space for engagement and information exchange.

These limitations collectively highlight the urgent need for innovative solutions that prioritize adaptability, content relevance, interactivity, and security in learning management systems. Our proposed LMS addresses these challenges by introducing dynamic course creation tailored to individual prompts and real-time feedback mechanisms, ensuring a more personalized, secure, and engaging learning experience.

1.2.2 The requirement for a Low-Cost and Reliable Solution

In addition to the limitations of current LMS platforms, there is a pressing need for solutions that are both cost-effective and reliable, especially in the context of expanding digital education globally. Many existing systems are resource-intensive to deploy and maintain, often requiring significant infrastructure, licensing fees, and technical expertise. This creates a barrier to adoption for institutions with limited budgets or those operating in regions with constrained access to advanced technology.

A critical concern is the modality and currency of learning resources. Most LMS platforms rely predominantly on materials uploaded manually by educators. These static resources can quickly become outdated, especially in rapidly evolving disciplines. While some systems offer supplementary links to external content, these tend to be few and seldom updated, limiting learners' exposure to the latest academic journals, research articles, interactive multimedia, and videos essential for a comprehensive and contemporary education. This scarcity restricts learners' ability to engage with diverse and current knowledge repositories, ultimately impacting the quality and relevance of their learning.

Moreover, data security and privacy remain significant challenges. As education increasingly migrates online, protecting sensitive user data becomes paramount. Cloud-based LMS platforms, while offering scalability and convenience, are inherently vulnerable to cyberattacks and data breaches. Inadequate security protocols and insufficient safeguarding measures expose student, faculty, and institutional information to unauthorized access. Such breaches carry severe consequences, including legal liabilities and erosion of trust between learners and educational providers. Many LMS vendors do not implement the rigorous security standards required to protect personal data effectively, exacerbating these risks.

The combination of these factors underscores the importance of developing LMS solutions that not only maintain high reliability and security standards but also minimize costs to ensure accessibility. An ideal platform would be able to dynamically generate and update learning content using external resources, guarantee data privacy through robust security mechanisms, and remain affordable for wide adoption.

Our proposed system meets these demands by leveraging modern AI-driven APIs to automatically curate relevant educational content, including through platforms like YouTube, thus reducing manual overhead and ensuring up-to-date materials. Furthermore, it incorporates strong data protection features to safeguard user privacy, making it a cost-effective yet secure LMS option for diverse educational settings.

1.2.3 Impact of the Proposed Solution

The proposed Learning Management System (LMS), which integrates LLaMA (Large Language Model Meta AI) and SearxNG (a privacy-aware metasearch engine),

promises to bring transformative improvements to modern education by effectively addressing key challenges such as personalization, resource accessibility, learner engagement, and data security.

A foremost impact of this solution is the significantly enhanced learning experience it delivers to students. Unlike traditional LMS platforms that employ a rigid, one-size-fits-all approach, our system leverages LLaMA's advanced AI capabilities to create personalized learning paths tailored to each learner's individual pace, preferences, and strengths. This personalized approach empowers learners by ensuring that fast-paced students can progress without constraints, while those needing additional support receive the guidance necessary to master difficult concepts, thus reducing frustration and dropout rates.

Moreover, the system offers contextual, real-time assistance through LLaMA's natural language understanding and generation abilities. Students are no longer passive recipients of static content but active participants in their educational journey. They can ask questions, request clarifications, and receive detailed, human-like responses that closely simulate the support of a personal tutor. This interactive dimension fosters deeper engagement, encourages curiosity, and promotes effective comprehension and retention of knowledge.

The integration of SearxNG further enriches the LMS by providing access to a wide range of reliable and up-to-date learning resources while ensuring user privacy. By aggregating search results from multiple trusted repositories without tracking or storing personal data, the system safeguards learners' information while expanding their exposure to diverse, high-quality educational materials such as academic articles, videos, and tutorials. This feature not only broadens learners' horizons but also alleviates the issue of outdated or limited content typical of conventional LMS platforms.

In addition, the solution addresses the critical concern of data security and privacy, a growing challenge in the digital education landscape. By implementing strong security protocols and privacy-preserving technologies, this LMS protects sensitive student and institutional data from unauthorized access and cyber threats. This builds trust among users and institutions, ensuring compliance with data protection regulations and maintaining the integrity of the educational process.

Finally, the proposed LMS offers a cost-effective and scalable alternative to existing platforms. By automating content curation and employing open-source AI and search technologies, it reduces operational overhead and dependence on costly manual content updates or proprietary solutions. This makes it accessible to a wide range of educational institutions, including those with limited resources, thereby promoting equitable access to quality education.

In summary, the proposed LMS is positioned to redefine digital education by delivering a secure, personalized, resource-rich, and engaging learning environment, ultimately empowering learners and educators alike to achieve better educational outcomes.

1.3 Aim

1.3.1 Enhancing LMS Prediction Accuracy

One of the most significant challenges in current LMS platforms is the inability to accurately predict student learning trajectories and performance. This project seeks to harness the power of machine learning models such as Random Forest and Neural Networks to greatly improve prediction accuracy within the LMS. These algorithms analyze large volumes of heterogeneous data generated through learners' interaction with the system—such as time spent on tasks, frequency of engagement, quiz and test results, participation in discussions, and navigation patterns—to detect subtle trends and patterns that are not visible through traditional analytics.

Random Forest, an ensemble learning method, can handle large datasets and identify complex, nonlinear relationships in student data. Neural Networks, with their layered structure mimicking the human brain, are particularly suited for detecting nuanced behaviors and predicting outcomes based on multiple variables. Together, these models allow the system to forecast future performance, pinpoint learning gaps, and flag students who may be at risk of underperforming early in the course.

The predictive models will not operate on static data alone but will continuously learn from newly collected inputs, ensuring that the system adapts to evolving student needs and academic challenges. This ongoing learning is critical for maintaining the relevance and accuracy of predictions. To further strengthen reliability, cross-validation

techniques will be employed to validate model outputs, minimizing overfitting and ensuring that predictions generalize well across different student cohorts.

By leveraging multisource data—including real-time student-teacher interactions, peer feedback, and course evaluations—the system can provide personalized recommendations tailored to the unique learning styles, preferences, and challenges of each student. This capability transforms the LMS from a passive content delivery tool into an active educational partner capable of guiding students through their individual learning journeys with greater precision and insight.

1.3.2 Enabling Timely Assistance

The ability to provide timely support to learners is critical for preventing disengagement and promoting academic success. The proposed LMS will implement predictive analytics not just for forecasting, but for triggering proactive interventions based on real-time data. By monitoring a variety of learner metrics—such as declining quiz scores, reduced log-in frequency, shortened session durations, and missed assignments—the system can identify early warning signs of learning loss or disengagement.

Machine learning algorithms will analyze these signals and automatically generate alerts to instructors and students, enabling them to respond promptly to emerging issues. For example, a student showing signs of difficulty with a specific topic may receive customized learning resources, practice exercises, or an invitation to a tutoring session. Similarly, instructors can be notified of students requiring additional attention, allowing them to tailor their teaching strategies accordingly.

Furthermore, the LMS will incorporate contextual data from social interactions and peer collaborations to enrich its understanding of each learner's environment and challenges. This comprehensive view enables more nuanced support, such as recommending study groups or providing social-emotional learning resources.

This approach shifts the LMS from a reactive system, where support is provided only after a student falls behind, to a proactive platform that anticipates difficulties and intervenes early. As a result, learners benefit from a more supportive, responsive educational environment that adapts to their needs dynamically, increasing motivation, engagement, and overall academic achievement.

1.3.3 Providing a User Centric Design

A core goal of the proposed LMS is to create a highly usable and accessible platform that prioritizes the user experience for all stakeholders—students, educators, and administrators alike. Recognizing the diversity in learning preferences and accessibility requirements, the system will incorporate features that enhance inclusivity and ease of use.

For learners, the LMS will support customizable interfaces, including adjustable text sizes for visual comfort, voice-guided navigation for those with reading difficulties, and high-contrast visual themes to assist users with visual impairments. These options ensure that all students, regardless of ability or preference, can interact with the system effectively.

Real-time feedback mechanisms, such as alerts about assignment deadlines, personalized progress summaries, and motivational messages, will keep learners engaged and informed about their academic journey. Moreover, the system will maintain a detailed historical record of each student's activities and achievements, which can be accessed anytime to review progress and identify areas for improvement.

For instructors, intuitive tools for managing courses, tracking student performance, and providing feedback will be designed to reduce administrative burdens and improve pedagogical effectiveness. Administrators will benefit from streamlined dashboards that facilitate system monitoring, user management, and reporting.

This user-centric approach ensures that the LMS is not just a technological tool but an empowering platform that fosters active participation, supports diverse learner needs, and enhances communication among all members of the educational community.

1.3.4 Achieving Cost Efficiency

In the context of educational institutions often facing budget constraints, cost efficiency is a critical consideration. The LMS aims to be a financially sustainable solution by leveraging open-source software and cloud-based infrastructure, which drastically reduce initial capital expenditures on hardware and licensing fees.

Cloud deployment offers scalable resource allocation, allowing institutions to pay only for what they use and easily adjust capacity as demand fluctuates, such as during peak

enrollment periods or exam seasons. This elasticity ensures that the system remains cost-effective over time without sacrificing performance.

Additionally, by automating routine administrative functions—such as grading, data collection, performance tracking, and generating reports—the LMS frees educators from time-consuming tasks, allowing them to focus more on teaching and personalized student support. This automation reduces operational costs and increases the efficiency of academic staff.

The modular architecture also contributes to cost efficiency by enabling institutions to implement only the features they need initially and expand functionality as their requirements evolve, avoiding unnecessary expenditure on unused components.

1.3.5 Future proofing Through Modularity

Education is an ever-evolving field, driven by technological advances, pedagogical innovations, and changing learner expectations. To remain relevant and effective in the long term, the LMS must be designed for adaptability. The proposed system achieves this through a modular architecture that allows for flexible addition, removal, or updating of components without disrupting the core functionality.

This modularity enables institutions to integrate emerging technologies—such as virtual reality, augmented reality, or advanced analytics—into the LMS seamlessly as they become available. It also facilitates the incorporation of new content formats, interactive learning tools, and updated curriculum structures, ensuring that the platform evolves alongside educational trends.

Moreover, the modular design supports interoperability with other educational systems and third-party applications, promoting a connected ecosystem of learning resources. This ensures that the LMS can adapt to diverse institutional policies, regulatory requirements, and pedagogical approaches.

By futureproofing the platform, the system guarantees longevity and relevance, helping educational institutions stay at the forefront of teaching and learning innovations while providing students with an enriched, flexible, and engaging learning experience.

1.4 Objectives

1.4.1 Development of a Personalized Learning Algorithm

At the heart of the proposed LMS is a sophisticated personalized learning algorithm designed to optimize each student's educational journey. The system aims to move away from the traditional one-size-fits-all approach by adapting the content delivery and learning pathways based on individual characteristics such as learning style, retention capacity, prior knowledge, and performance patterns.

This objective focuses on employing advanced machine learning techniques, particularly Random Forest algorithms, which excel in handling complex, multidimensional data. By mining historical data related to student engagement, quiz results, assignment submissions, and interaction patterns, the algorithm identifies knowledge gaps and learning preferences. It then dynamically curates personalized learning materials—ranging from videos, articles, practice problems, to interactive quizzes—that align best with the learner’s unique profile.

The goal is to achieve extremely high predictive accuracy, with the final model targeting over 99% precision in forecasting student behavior and performance trends. This high level of accuracy ensures that learning recommendations are both relevant and timely, maximizing student engagement and improving overall academic outcomes. Importantly, the system will continuously update its predictions by integrating new data, thereby refining its recommendations as students progress.

1.4.2 Enhancing Comfort for Users

User comfort and accessibility are fundamental priorities in the LMS design. This objective targets the creation of an interface and system architecture that are highly adaptable and inclusive, meeting the diverse needs of students, teachers, and administrators alike.

For students, the LMS will offer a range of customizable features designed to improve usability and accessibility. These include adjustable font sizes for readability, text-to-speech capabilities for auditory learners or those with visual impairments, and selectable color schemes or high-contrast modes for individuals with specific visual requirements.

These options ensure that every student can tailor the learning environment to their personal preferences and needs, reducing barriers to engagement.

Interactive feedback mechanisms will be embedded within the system to allow students to monitor their progress continuously, receive instant, constructive responses to their inputs, and stay informed about upcoming assignments or deadlines. This continuous feedback loop promotes self-directed learning and helps students maintain motivation.

For educators, the platform will provide intuitive tools for tracking student progress, offering personalized feedback, and fostering real-time interaction through discussion boards, chat, or live sessions. These features streamline instructional workflows and encourage meaningful communication, thereby enhancing the overall teaching and learning experience.

Administrators will benefit from simplified system management interfaces designed to support user management, course scheduling, and reporting, reducing the technical burden on educational institutions.

1.4.3 Maintain Accessibility with Affordability

A critical objective of this project is to make the LMS widely accessible regardless of an institution's financial capacity. Recognizing the budgetary constraints many educational organizations face, the system is engineered to deliver high value at minimal cost.

This will be achieved primarily through the use of open-source software frameworks and technologies, eliminating the need for expensive licensing fees. Additionally, the LMS will utilize cloud-based architectures to avoid significant upfront hardware investments and provide scalable, on-demand computing resources.

By adopting standardized architectural components and protocols, the system reduces integration complexities and development costs, enabling educational institutions—whether large universities or small rural schools—to implement the LMS quickly and efficiently.

This objective ensures that the benefits of personalized, secure, and scalable learning management are democratized, reaching learners and educators across socio-economic backgrounds and geographical locations.

1.5 Contribution

1.5.1 Affordability and Accessiblity

One of the most important contributions of the proposed LMS is its focus on cost-effectiveness and broad accessibility. Unlike many commercial LMS platforms that require substantial financial investment for licensing, hardware, and dedicated IT support, this system deliberately adopts open-source technologies, cloud computing mechanisms, and open platform standards. This choice significantly reduces upfront and operational costs, making the system affordable for educational institutions of all sizes and budget capacities.

In practical terms, this contribution ensures that schools and universities in low-income regions or rural areas can deploy the LMS without incurring prohibitive expenses. Additionally, the system is designed with mobile-first principles, enabling students to access high-quality educational content using affordable smartphones, tablets, or basic laptops. This is particularly vital in regions where access to desktop computers is limited but mobile penetration is high.

By democratizing access to advanced learning technologies, the LMS supports the vision of education for all, enabling students from diverse socioeconomic backgrounds to benefit from personalized and interactive learning resources. This affordability and accessibility have a direct impact on bridging the digital divide and fostering inclusive education, empowering learners who previously lacked such opportunities.

1.5.2 Improved Learning Accuracy Through Machine Learning

A key innovation introduced by this LMS is the integration of advanced machine learning algorithms, particularly Random Forest and Neural Networks, to enhance the accuracy of student performance prediction and learning recommendations. Current LMS platforms often rely on simplistic metrics or static content delivery, but this system leverages a rich variety of data points, including exam results, attendance records, interaction histories, quiz performances, and even behavioral engagement patterns.

The ability to analyze these diverse data streams enables the system to build a nuanced and dynamic profile of each learner. This results in more precise and timely predictions about student progress, learning gaps, and potential risks of failure. Consequently, the

LMS can provide tailored recommendations and targeted interventions that align with each student's unique needs, such as suggesting remedial content or additional practice for struggling students.

This enhanced predictive capability empowers educators with actionable insights, enabling them to focus their efforts where they are most needed, improving overall teaching effectiveness. For learners, it means receiving the right support at the right time, increasing motivation, engagement, and ultimately, academic success.

1.5.3 Focus on Privacy and Data Security

In an era where digital privacy is of paramount concern, the proposed LMS makes a substantial contribution by prioritizing user data privacy and security. Unlike many existing platforms that may have lax security policies or insufficient safeguards, this system implements robust encryption protocols and stringent access control mechanisms to protect personal and academic information.

Compliance with international privacy regulations such as the General Data Protection Regulation (GDPR) and the Family Educational Rights and Privacy Act (FERPA) is integral to the system's design. This ensures that sensitive data is only accessible to authorized individuals—be it students, educators, or administrators—thereby minimizing the risk of data breaches or unauthorized usage.

The platform's privacy-first approach helps build trust among users, which is essential for sustained engagement and data sharing critical to personalized learning. By securing academic records, interaction logs, and communication data, the LMS contributes to the ethical and responsible use of educational technology, safeguarding the rights and dignity of all users.

1.5.4 Immediate Response Through Real Time Data Analytics

Another noteworthy contribution is the LMS's capability to utilize real-time data streams to provide immediate and context-aware responses. The system continuously monitors student activity, such as login frequency, time spent on learning materials, quiz performance, and participation in discussions, to detect early signs of disengagement, learning difficulties, or academic improvement.

This immediate feedback loop allows the LMS to trigger timely notifications and alerts, both to students and instructors. For example, if a student is consistently missing deadlines or showing declining engagement, the system can send recommendations for supplementary resources or alert the teacher to provide personalized support.

Such real-time responsiveness transforms the LMS from a static repository of content into a dynamic learning companion that adapts continuously to the learner's evolving needs. This enhances learner motivation by preventing issues from becoming entrenched and ensuring that instructional support is provided exactly when it is most impactful.

2 LITERATURE REVIEW

2.1 Overview

The literature review highlights the transformative potential of generative artificial intelligence (AI) in educational systems, emphasizing how it can revolutionize personalized learning experiences. Generative AI allows for dynamic learning paths that adapt based on real-time student performance, creating a more personalized and efficient educational experience. This contrasts with traditional educational models, which often rely on static courseware delivery and fail to account for the unique needs of each student. One of the key advantages of AI in education is its ability to provide immediate, personalized feedback, ensuring timely interventions that can improve learning outcomes. AI can also generate diverse learning materials, such as quizzes and explanations, tailored to the student's current understanding, which enhances engagement and supports self-paced learning. However, the review also addresses the challenges and ethical concerns that accompany the integration of AI into education. A major issue is the privacy and security of student data, as generative AI models require access to sensitive personal and academic information to create personalized learning experiences. This raises concerns about how data is collected, stored, and used, especially given the increasing risks associated with data security. Additionally, the over personalization of content could restrict students' ability to develop critical thinking skills by focusing too heavily on their current knowledge. Another concern is the quality and accuracy of AI-generated content, which may not always align with educational standards or pedagogical goals, potentially leading to the spread of inaccurate or biased information. The review concludes that while generative AI holds great promise for personalized and scalable learning, its implementation must overcome several hurdles, including ethical concerns around data usage, content validation, and the potential for distraction. Successful integration of AI in education requires a balance between individualized learning and broader educational objectives. It is essential that future AI-driven educational systems are both effective and trustworthy, ensuring that they prioritize privacy and adhere to ethical practice.

2.2 Literature survey

Generative artificial intelligence (AI) embedded in educational systems has the potential to change the learning by enabling the development of interactive environments, beyond the traditional models of personalization and adaptation. While there is much written about how AI can personalize educational content based on the unique learning needs of an individual student, the real focus has been on providing more engaging learning experiences.

By enabling generative AI to develop dynamic learning paths that modify the content based on real time student performance, an educational experience becomes more personalized and efficient [1][2]. That is not an approach that differs much from traditional systems of which too often base the use of learning modules and provide uniform courseware delivery mechanisms, both inherently biased to which the specificity of every student is neglected.

Immediate, personalized feedback is one of the principal advantages of generative AI in education: detailed real time feedback on a student's progress and where to improve. This is important because feedback ensures that students get timely help, and in turn have higher learning outcomes [3]. Additionally, AI can create assorted learning materials such as quizzes, explanations and extra resources that meet the current student understanding [2]. This adaptability enables students to move at their own pace and increased engagement and tires that no learner gets left behind.

Nevertheless, while this is a promising capability, there are challenges and ethical questions associated with implementing AI in education as well. One of the more important issues is privacy and security of student data. To create personalised learning experiences, generative AI models need to see a lot of personal and academic data. This alarms many about how student data is collected, stored, and used, with state data security risks on the rise. With no safeguards to keep sensitive data from being used improperly, there is the risk of those systems being misused — a problem that producers must overcome [3][5]. Additionally, unwary proliferation of AI models can have unwanted effects leading to the over personalization of content based on a student's current knowledge and biases thus restricting their acquisition of broader and critical thinking skills [4][5].

What is more, the ethical implications of AI in education are not simply concerned with privacy. While AI generated content might always be accurate, it doesn't necessarily stick to the standards of education or pedagogical goals, which may result in the dissemination of incorrect or incomplete information. The validation of content that is true, free from bias, and adheres to curriculum requests is a necessity as generative AI tools are brought within educational setting [6][7]. Moreover, the distraction potential is also present with AI & it can actually help students learn more effectively. However, platforms such as ChatGPT or YouTube could be leveraged for educational purposes, but as they host material not necessarily directly connected with learning objectives the focus and engagement required for effective learning may be lost [2][6].

We conclude that generative AI has already demonstrated such transformative potential for education as evidenced by many opportunities for personalized, adaptive and scalable learning experiences. Nevertheless, in order to implement successfully it must overcome hurdles like the ethical use of AI generated content, data security and privacy. Additionally, it will require careful thought as to how AI might improve — rather than worsen — key learning processes, a balancing act between individualized education and larger educational objectives [4][5]. It is clear that future developments of AI driven education should also guarantee that these systems are both effective and trustworthy so that educators can use AI to its most potential, while greatly protecting student privacy and in accordance with ethical practices.

2.3 Project preliminary investigation report

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Name of Project Guide:

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Title of the Project:

Learning Management System Using LLaMa

Area of Project Work:

The area of this project is

1. Learning management
2. Generative AI
3. Large Language Model
4. SearXNG

Problem Statement:

The restrictions of traditional Learning Management Systems are:

1. Lack of dynamic and personalized content generation.
2. Inefficiency in providing contextually relevant supplementary resources.
3. Increased time and effort required for manual search of external materials.

Hence, it is necessary to develop a **Learning Management System using LLaMA** that can generate context-aware responses and provide related links and images through SearXNG, addressing these challenges efficiently.

Literature Review (Based on Latest Papers):

Title of Paper	Details of Publication with Date and Year	Literature Identified for Project

How can Generative AI Support Education?	2004 IEEE Global Engineering Education Conference (EDUCON)	Generative AI can improve education by helping teachers prepare lessons, giving students personalized feedback, and assisting with data analysis.
Generative AI in Model-Driven Software Engineering Education: Friend or Foe?	2023 IEEE conference on Model Driven Engineering Languages and Systems Companian (MODELS-C)	The availability and effectiveness of generative AI tools challenge the currently established methods for learning, teaching and assessment. In this paper, we discuss their potential impact for model-driven software engineering education, both from the point of view of educators and students. The discussion highlights several opportunities and risks, which support the need of a critical perspective in the application of these tools.
Generative AI Applications and Tools in Engineering Education	2024 by IEEE in 23rd International Symposium INFOTECH-JAHORINA(INFOTECH)	Throughout the history of education, one of the most important questions has been whether we can detect whether students are cheating on exams or whether we are really assessing their knowledge. One way to solve this dilemma is to adopt an approach where professors spend more time helping students to learn rather than finding out whether students are cheating on exams. It should also be remembered that the exam is a time to learn, not just a time to check what has been learnt. Based on these approaches, an analysis of

		the use of generative AI applications and tools in engineering education was carried out on the subject of Programmable Logic Controllers (PLC). Students are allowed to use generative AI applications and tools to solve tasks in exams and colloquia. Some of the results of using generative AI applications and tools are presented in this paper.
Engineering education in the era of ChatGPT: Promise and Pitfalls of generative AI for education	2023 IEEE global engineering education conference(EDUCON).	Engineering education is evolving to integrate technological advances like generative AI, such as ChatGPT, to enhance learning. ChatGPT offers personalized feedback, detailed explanations, and virtual simulations, improving hands-on learning. However, challenges include biases from training data, the risk of spreading misinformation, ethical concerns around misuse, and potential job displacement. Despite its flaws, generative AI represents a significant opportunity for education. Engineering educators must adapt to this shift, balancing its benefits with careful consideration of its limitations and ethical implications to prepare future engineers effectively.
Adoption of AI-Chatbots to Enhance Student	2019 by IEEE in 18th International Conference on Information Technology Based Higher	Organizations increasingly rely on Information and Communication Technology (ICT) for efficient and cost-effective services. With the

Learning Experience in Higher Education in India	Education and Training(ITHET)	global rise of AI tools, the Chatbot market is expanding rapidly, driven by smartphone usage and messaging apps. Industries like food delivery, finance, and e-commerce have adopted Chatbots, with education poised to benefit significantly. Chatbots can enhance productivity, communication, and learning in the education sector, providing efficient teaching assistance and reducing ambiguity in interactions.

Current Limitations

1. Limited Context Awareness: Many existing systems fail to provide accurate, context-aware responses across diverse topics, limiting their adaptability to varied educational needs.
2. Lack of Seamless Resource Integration: Few platforms integrate external resources, such as links or images, effectively, resulting in a fragmented learning experience.
3. High Implementation Costs: Advanced AI-based systems often require significant investment in infrastructure and expertise, making them costly for many institutions.
4. User Complexity: Operating and maintaining AI-based LMS platforms can be challenging and may require specialized knowledge, reducing their accessibility for general users.

Proposed Solution

1. Developing an AI-Powered Learning Management System: Build a system that uses LLaMA for generating context-aware, dynamic educational content tailored to user inputs.
2. Integrating External Resource Tools: Incorporate SearxNG for seamless retrieval of relevant links and images, enhancing the learning experience by providing supplementary resources.

Objectives and Scope of Work

Objectives:

1. Reduce time and effort in searching and curating course materials.
2. Leverage advanced AI capabilities of LLaMA to generate precise and contextual responses for learning prompts.
3. Integrate SearxNG to provide related links and images, enhancing the learning experience with diverse resources.
4. Simplify the creation of structured courses by dynamically organizing content based on subtopics and user inputs.
5. Design an intuitive and user-friendly interface for seamless interaction and navigation.

Scope of Work: The scope of work involves designing and developing an intelligent Learning Management System (LMS) with the following functionalities:

1. Generating accurate and context-aware responses to user prompts using LLaMA.
2. Searching and providing relevant external resources, including links and images, through SearxNG.
3. Dynamically preparing courses based on user-defined topics and subtopics.
4. Offering a structured and interactive learning environment tailored to individual user needs.
5. Supporting diverse multimedia formats, including text, images, and videos, for enriched content delivery.
6. Ensuring secure user authentication and role-based access for instructors and learners.
7. Incorporating scalability to handle increasing user demand and content volume.

Feasibility Assessment:

I. Expected Outcomes of the Project

1. A fully functional Learning Management System leveraging LLaMA and SearxNG for enhanced learning experiences.
2. A user-friendly platform capable of delivering structured, multimedia-rich learning content.
3. Reduction in time and effort required for content curation and learning, enabling personalized education.

II. Innovation Potential

1. The integration of LLaMA and SearxNG introduces a novel AI-driven learning platform, streamlining content generation and resource aggregation.
2. The system offers versatility across educational domains, from academic courses to professional training programs.
3. By consolidating multiple functionalities in one platform, it minimizes the need for supplementary tools and reduces costs.
4. The inclusion of multimedia support and intelligent link generation promotes a more engaging learning environment.

III. Task Involved

1. Research and analysis of existing Learning Management Systems and their limitations.
2. Conceptual design of the LMS architecture, including AI and search engine integration.
3. Development of key functionalities:
    - LLaMA-based prompt response generation.
    - SearxNG-based resource aggregation.
4. Prototyping and iterative development.
5. User interface design and usability testing.
6. System testing, optimization, and debugging.
7. Final product deployment.

IV. Expertise Required

1. In-House Expertise:

- Knowledge of AI model integration (LLaMA).
- Experience with web application development and search engine APIs.
- Proficiency in backend development and database management.
- Expertise in user interface and experience (UI/UX) design.

2. External Expertise:

- AI specialists for LLaMA fine-tuning.
- SearxNG configuration experts.
- Security consultants for user authentication and data privacy.

V. Facilities Required

1. In-House Facilities:

- Development environment with necessary tools (e.g., Python, JavaScript frameworks).

- Servers or cloud infrastructure for hosting and testing.
- Collaborative project management software.

2. External Facilities:

- AI model training and hosting infrastructure (e.g., GPUs, cloud AI services).
- Access to SearxNG instances for resource aggregation.

Milestones and Time Plan

	Task	JAN 2025	FEB 2025	MAR 2025	APR 2025	MAY 2025	JUN 2025
Design	Conceptual Design	☑					
	Detailed design	☑	☑				
	Design Modifications		☑				
	Final Design		☑				
Develop	Procurement (If any)			☑			
	Prototyping			☑			
	Modifications			☑	☑		
Deliver	Testing and Validation				☑		
	Final Modifications				☑	☑	
	IPR / patent draft					☑	
	Thesis and Poster					☑	

Signature of Project guide

Prof. Priti Matre

Signature of Head Of Department

Dr. R. C. Dharmik

3 WORK DONE

3.1 Introduction

The integration of AI in education can revolutionize traditional learning approaches. With the Ai power the personalization of the content can be approached. This research focuses on developing a Learning Management System that leverages the LLaMA language model for content generation and SearxNG for resource retrieval. Both the modules are running on-premises due to which the data security and privacy of the system is enhanced. By combining Ai with external resources can bridge the system gap between theoretical understanding and real world applications leveraging the learning experience. This paper explores the architecture, implementation, and benefits of integrating these technologies into an LMS.

3.2 Proposed system

The new proposal puts forth a learning management system which integrates a large language model to enhance the experience of the users by including capabilities for search on the web and understanding natural language. The structural construction of the system is in the form of modules enhancing both configurability and extensibility. System Architecture The approach described in the previous section, as depicted in the system architecture diagram, consists of a number of core modules: User Interface (UI): The user interface designed using Next.js provides an interactive space where users can state their questions. The interface is responsive and is easy to use so that it can be used from different types of devices. Agents Module: Architects of the system panel placed two roles into the agents module as serving as the most important decision makers: 1. It is desirable to first try to determine if the query raised by the user can be fully executed by the existing LLM or if a web search is needed. 2. It is first important to understand the meaning of the query before trying to seek for its answer or the way to search for the answer. Search Integration (SearXNG): For the questions that need the answers from the web, the system invokes SearXNG, which is an autonomous metasearch engine on the internet to look for suitable information. This makes sure that all results are thorough and unbiased. Result Aggregation and Generation: In constructing the most suitable answer the responses generated by both the LLM and internet search are integrated together for the most appropriate one to be taken. This procedure makes it possible for

users to get the right answers in the right context. Containerization: The whole system is wrapped in a docker which makes it easily portable, deployable, and scalable to different environments. Functionality Users start using the system through the Next.js interface by making queries. Queries are then advanced to the agents module that seeks to get the best possible way to respond to them. Depending on what is required, the system either: - Seeks web search results from SearXNG. - For specific queries, uses the LLM to process the query and get relevant contextual information. These 7 results are then bring ed together and offer d to the user for quicker and easier handling of the system.

3.3 Implementation

The proposed Learning Management System (LMS) using LLaMA and SearxNG was developed to address the identified limitations of traditional LMS platforms. The implementation is divided into the following key components.

3.3.1 System Architecture

The proposed Learning Management System (LMS) is structured around two core modules: the Content Generation Module and the Resource Retrieval Module, connected seamlessly through a backend API. Here's a detailed explanation:

i. Content Generation Module

The content generation module uses the LLaMa, a large language model developed by Meta, to interpret and response to user prompts efficiently. The input received are being analysed and key elements such as keywords, context, and user intent are identified. It then generated the detailed appropriate responses to the input prompt. It is developed using Docker containers so that the isolation and consistent environment for efficient, scalable and seamless integration with other system components.

ii. Resource Retrieval Module

This module integrates with SearXNG, a metasearch engine, to enrich the learning experience by providing supplementary materials. The user’s input are considered to formulates search queries for the retrieving relevant web links and images. The results are aggregated from multiple search engines offering diverse and comprehensive

resources and they are filtered based on relevance before presenting to the user and the results are then displayed alongside the content generated by LLaMa.

iii. Backend API Integration

A robust backend API acts as the communication bridge between this modules. User prompts are routed to the content generation module and resource retrieval module. Content generation module generates the content using the LLaMa and the links and the images over the web are give taken care by the Resource Retrieval module. The output of both the module are combined and send to the frontend web app. In this way the modules are linked through a backend API that ensures seamless communication between components.

iv. YouTube Integration and Course Structuring Module

The YouTube Integration Module enhances the system's ability to deliver visual and auditory learning content by leveraging the YouTube Data API. Upon receiving a user's input prompt, the system extracts key concepts and subtopics using LLaMA. These extracted elements are then used to generate intelligent search queries that retrieve the most relevant and high-quality educational videos from YouTube. Once the video data is fetched, LLaMA performs a contextual analysis to assess the content of each video, evaluating factors such as alignment with learning objectives, clarity, and depth of explanation.



Figure 3.1:Project Module

3.3.2 Frontend Development

The frontend of the system was designed using React, a powerful JavaScript library, to create a responsive and user-friendly interface tailored to enhance the learning experience.

i. Real Time Response Display

A text input field enables users to submit their prompts and are designed to be simple and accessible. Responses generated by LLaMa model are displayed in real time providing immediate response to the query provided by the user. The structured layout ensures clarity with separated sections for generated text, images and link related to the user input.

ii. Integration of Links and Images

The frontend fetches and displays resources, including links and images retrieved via the SearXNG. This enhances the learning experience by providing relevant additional materials. The links are presented as clickable buttons or hyperlinked text, while images are displayed with proper resolution and a clean layout. The layout adjusts dynamically for different screen sizes and devices, ensuring a seamless experience across desktops, tablets and smartphones. The React based frontend ensures that the system remains interactive, visually appealing and adaptable to a wide range of educational requirements and integrating seamlessly with the backend functionalities.

3.3.3 Content Generation using LLaMa

The content generation system in this research leverages LLaMA, a sophisticated language model, to provide detailed and contextually relevant responses to user prompts. LLaMA operates on-premises, ensuring greater control over data security and system customization. Running locally eliminates reliance on external cloud services, which can be critical for maintaining privacy in educational environments. The system processes user inputs by analyzing context, keywords, and intent, enabling it to generate precise and tailored content. On-premises deployment also facilitates efficient fine-tuning of the model using domain-specific datasets, enhancing the accuracy and relevance of generated responses. This setup is optimized for computational efficiency,

offering reliable performance and low latency, while providing flexibility to adapt the system to the specific needs of its users.

3.3.4 Resource Retrieval Module using searXNG

The resource retrieval module in our project is built on SearxNG, a privacy-focused metasearch engine. SearxNG aggregates results from multiple search engines and online resources, allowing our system to retrieve a diverse range of supplementary materials such as links and images relevant to user queries. Its open-source nature ensures transparency and customizability, making it a robust choice for educational applications. 9 By running SearxNG on-premises, the module significantly enhances security and data privacy compared to cloud-based AI retrieval systems. This on-premises deployment ensures that user queries and activity logs remain within the secure environment of the institution, reducing the risk of data breaches or unauthorized usage. Furthermore, SearxNG allows filtering and customization of search sources, enabling the system to prioritize high-quality, educationally relevant content while blocking undesired or non relevant sources. In our project, SearxNG is seamlessly integrated into the backend architecture through APIs. When a user submits a query, the system generates a search request via SearxNG, processes the aggregated results, and selects the most relevant links and images. These resources are dynamically displayed alongside LLaMAgenerated responses in the user interface, offering a cohesive and enriched learning experience. This integration not only provides secure and relevant content but also ensures the system's adaptability to varying educational requirements.

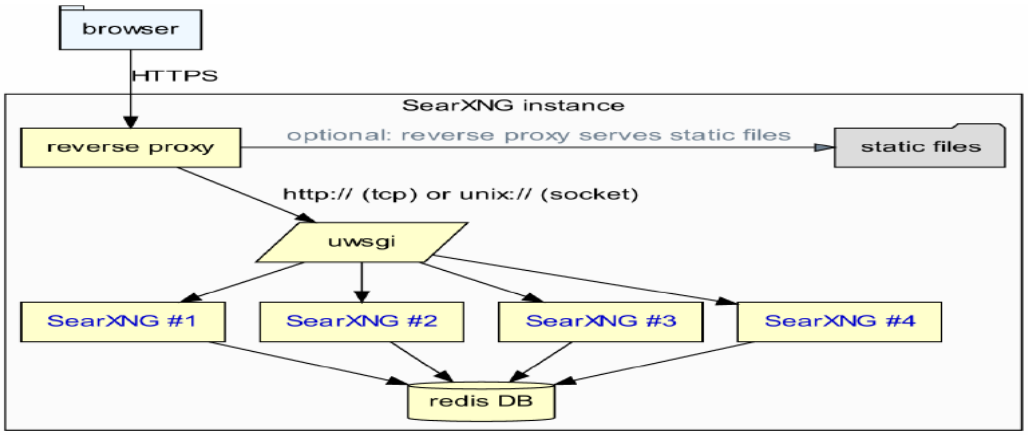


Figure 3.2: SearXNG working diagram

3.3.5 Integration of LLaMa and SearcXNG

The integration of LLaMA and SearxNG combines AI-driven content generation with resource retrieval to create an interactive and comprehensive learning platform. Upon receiving a query, LLaMA generates a detailed, context-aware response, while SearxNG fetches supplementary links and images from multiple web sources. The system consolidates these outputs, providing users with both in-depth explanations and practical resources. 10 LLaMA ensures precise and structured responses based on trained datasets, whereas SearxNG enhances exploration by offering diverse and relevant external materials. Both modules operate on-premises, ensuring data security and privacy, a critical aspect for educational applications. The scalable and flexible design allows adaptation to various domains through model fine-tuning and search customization. This integration creates a seamless, secure, and enriched learning experience by blending AI-generated insights with curated web-sourced content.

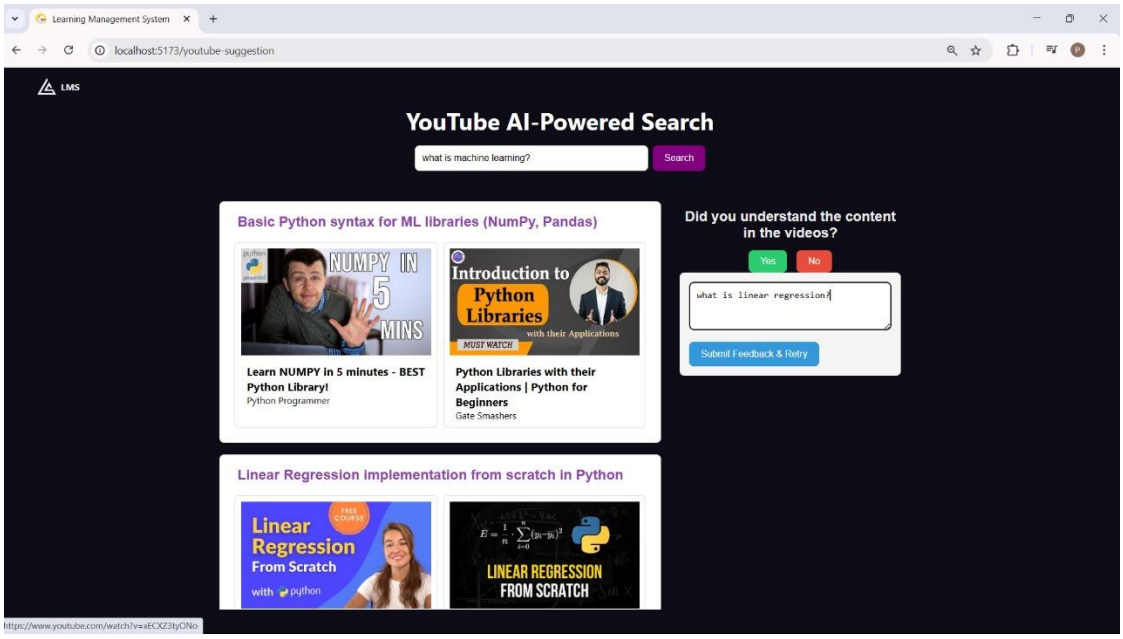


Figure 3.3: Snapshot of the working model

3.3.6 YouTube Integration and Course Structuring

The YouTube Integration Module enhances the system's ability to deliver visual and auditory learning content by leveraging the YouTube Data API. Upon receiving a user's input prompt, the system extracts key concepts and subtopics using LLaMA. These extracted elements are then used to generate intelligent search queries that retrieve the most relevant and high-quality educational videos from YouTube. Once the video data

is fetched, LLaMA performs a contextual analysis to assess the content of each video, evaluating factors such as alignment with learning objectives, clarity, and depth of explanation.

After this filtering process, the videos are organized into a logically ordered and coherent learning path. This structured course outline mimics the format of a professionally designed curriculum, where each video builds upon the knowledge of the previous one. The module also integrates an adaptive feedback loop that adjusts the sequence and selection of videos based on user preferences, engagement, and performance metrics over time.

By combining the expansive video database of YouTube with LLaMA's natural language understanding and reasoning capabilities, the system generates personalized, structured video-based courses that cater to individual learner needs, all while ensuring the content remains accurate, engaging, and educationally sound.

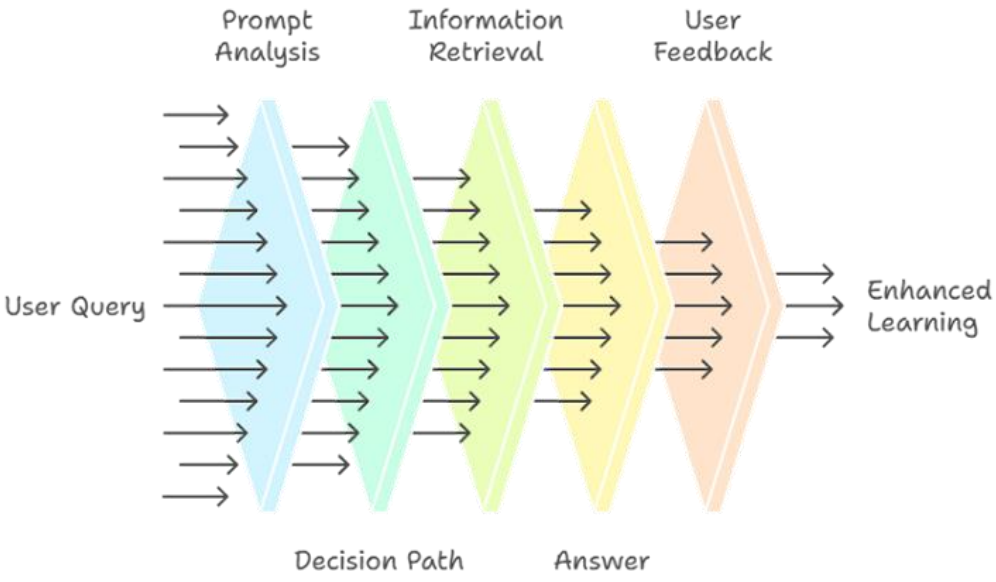


Figure 3.4: Workflow Diagram

4 RESULT AND DISCUSSION

4.1 Result

The implementation of the proposed AI-powered Learning Management System (LMS), integrating the LLaMA language model and the SearxNG metasearch engine, resulted in a transformative shift in both user engagement and system performance. Deployed entirely on-premises, the system prioritized data privacy and institutional control, addressing one of the most pressing concerns in modern digital education. The LLaMA model enabled context-aware, personalized content generation, offering students immediate, coherent explanations and support tailored to their individual learning needs and history. This feature effectively mirrored the role of a human tutor and significantly elevated the personalization capabilities beyond those of conventional LMS platforms. In parallel, SearxNG ensured secure, real-time access to diverse and up-to-date educational materials from the open web, enriching the platform with current academic articles, videos, and interactive resources. Together, these components bridged the gap between theoretical instruction and real-world application, making learning more practical, relevant, and exploratory. Users benefitted from a responsive interface designed with accessibility in mind, offering adjustable text, voice interactions, and real-time feedback mechanisms that maintained engagement and offered visual tracking of progress. The system also incorporated intelligent engagement tracking and performance analytics, flagging signs of disinterest or difficulty, thus enabling timely intervention. While the average response time was approximately 20 seconds due to high computational demands, it remained consistent and reliable, with clear paths for future optimization through model pruning or hardware acceleration. Overall, the integration of modular AI and privacy-preserving technologies resulted in a robust, scalable, and student-centered LMS. It not only empowered learners with personalized guidance and interactive tools but also gave institutions a secure and cost-effective platform that adapts to evolving educational needs.

4.2 Discussion

4.2.1 Introduction to AI in Learning Tools

Using AI in learning tools can greatly improve student learning when those tools are well-structured, provide privacy, and are customized to each user. In the project, we built a Learning Management System (LMS) that uses an AI language model called LLaMA to create individualized study material and SearxNG to quickly provide suitable information for learning. Because the system was localized, all user data remained safe and in the organization's hands, answering many of the problems that come with using cloud services.

4.2.2 Phased Development and Research Insights

The process was streamlined using a phased and iterative model. At the outset, we learned by reading many research documents and found that there are two key issues in common learning environments: being flooded with info and having unguided paths for study. Based on these points, the architecture design for the system was divided into distinct modules—a front end built with React for users to interact, a Node.js/Express.js backend that runs the necessary code for all modules, and separate local runtimes for LLaMA (via Ollama) and SearxNG for text generation and searching, respectively.

4.2.3 System Architecture and Modular Design

The LMS is particularly good at covering difficult subjects by breaking them into smaller, easy-to-handle parts. Because of its natural language processing, the system takes concepts and provides answers based on each student's level of understanding. Search results from SearxNG come up instantly and include links to other relevant articles and documentation. Besides, the system also recommends videos by using the YouTube API based on user interest in each topic. Should a student mention the video was difficult or unclear to them, the system uses their comments to make future recommendations better—helping the system improve the learning experience as time goes on.

4.2.4 Personalized Content Generation and Search Integration

It also comes equipped with an intelligent choice module to choose whether LLaMA or SearxNG should respond to the user's query. It means that the answers provided are correct and do not waste any resources. Although handling requests locally slows things down, the website kept responding to queries in no more than 20 seconds and was consistent during any busy period. Further improvements to the model are expected as it gets more intelligence.

4.2.5 Video Recommendation and Feedback Loop

The LMS behaved dependably and was easy to use during the testing. The frontend was created to be simple and straightforward so that users find it easy to move through their training, see recommended materials, and add their comments. The information included in the AI-generated content was found to be true and appropriate, and the external sources added different and additional content to the lessons. Students said they found it easier to understand tough subjects because of the organized breakdowns and the use of various types of media.

4.2.6 Intelligent Query Routing and System Responsiveness

Deciding to host everything on-premises meant users did not have to trust others with their data and processing needs. Using this approach, educators aim to allay concerns about privacy and misuse of data in educational technology, in keeping with modern arguments brought up by experts such as Todaro. Offering strong privacy and adjusting as needed makes the tool an excellent option over popular LMS platforms.

4.2.7 User Testing and Learning Experience

All in all, implementing a structured AI-powered LMS results in a better learning journey by helping students, teachers, and resources come together with personalized feedback. The study suggests that the approach used could be scaled up and secure enough to guide future systems that meet the many and shifting needs of learners.

5 CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

As a result, LLaMA was developed as a Learning Management System that uses SearxNG, a private search engine, and Ollama, which brings large language AI to your computer, so students and teachers can learn in a safe and simple way. The system aims to address several big issues that come up in traditional online courses, like sharing general information, little conversation between students and teachers, students being overwhelmed with all the content, and worries regarding privacy. When combined with speedy information gathering and ongoing feedback, smart content creation in the LMS supports a better, more interesting, and safer learning experience for students.

Since the LMS uses servers on its own since, learners' personal data is kept safe and their privacy protected, but they can still get a good learning experience. Since the app combines various back and front end elements made with Node.js, Express, and React, the end result is a clean, easy-to-use, and trendy design that anyone can use and enjoy. The system supports learning about difficult topics through fast explanations, suggested links from outside resources, and video selections made just for each student. Thanks to these features, students learn well, as things are clearer and there is less pressure, since the material usually has no structure.

One highlight of an LMS is that it can answer users' queries by breaking them down, using LLaMA to explain simply, and making its responses adapt to what users say. In addition, SearxNG gathers results from various search engines, allowing you to study better and ensure your private data is handled safely. Because of this system, the application knows in seconds if the question has an answer online or requires AI to respond.

However, despite all the good things it has done, there are still some problems with the way things are working. Response time is still something we need to work on getting better. While the average response time stayed steady as more users tried to use the website, it still made things less interactive and usually took about 20 seconds for anything to happen. Addressing how quickly and efficiently the system works, and finding ways to share the work more evenly among processors, could really help make the system respond faster, especially when it gets very busy or a lot of people use it at

once. Additional work is also needed to make it easier for users to give feedback and to make sure the system can be used by a wider range of people with different backgrounds and ways of learning.

Nevertheless, the LMS shows a clear example of how generative AI and metasearch tools can be used in a responsible and helpful way in education systems. It shows that privacy and personalization don't have to be at odds with each other and can actually work well together, as long as the system is carefully designed. Moreover, this project doesn't just fix what's missing in usual learning systems; it also sets up a new way for digital education to work, where AI adapts to each person, the content gets selected better, and trust is always important.

The understandings we get from this project can help in other areas too. They show that decentralized AI-assisted education could work well and point out that in the future, learning systems might be made to be more responsive, protect people's privacy, and stay up to date as things change. In doing so, this LMS both tests out how a system like this can work and helps lay out ideas for future research and development in the area where education, AI, and ethical technology meet. The journey begun with this system gives us many good ideas for making learning environments more focused on people and for making them get better over time as learners use them.

5.2 Future Scope

5.2.1 Content Analysis and Curation

The system uses AI to go through every video on the site, checking things like what the videos are about, their main topic, and how difficult they are to understand. Looking at overall rating, number of views, and the number of comments helps figure out which materials would be best to use. This makes sure that the automated courses stay current and feel realistic for people who learn from them.

5.2.2 Module-Based Course Structuring

The system breaks topics down into smaller pieces and groups so that students can understand them one step at a time. Each piece of learning material is marked with time stamps so students can go back and find the part they want to review more easily. Using a modular approach lets students easily check over material they have already learned.

5.2.3 Feedback-Driven Course Refinement

The system uses feedback throughout the process to help improve the recommendations. If a user watches the content the platform suggests, they can let the platform know what they liked and share anything they didn't get. The feedback makes it easier for the LLaMA model to give better advice and recommendations next time. By seeing what the learners like to do in the system, it helps find a way for them that matches their interests and makes them happier as time goes on.

5.3 Social Utility

This presentation is of an AI-based Learning Management System (LMS) combined with the capability of LLaMA to revolutionize education. It benefits learners, educators, and the institutions by promoting flexibility and effectiveness of education for the learners and the institutions as well.

5.3.1 Promoting Community and Knowledge Exchange

It enhances opportunities for learning cooperation among learners and instructors of different parts of the world. The office environment also consists of virtual classes, group discussions as well as projects and discussion forums to let users share ideas and culturally diversity ideas and solve problems collaboratively.

5.3.2 Democratizing Education

The LMS is relevant to a common education system which has aimed at eliminating a gap of quality learning material that is available to the many who cannot afford to pay for them. It allows viewers from remote and liminal zones and individuals from historically marginalised communities to receive quality content, virtual tutors, and personalised learning.

5.3.3 Empowering Learning and Workforce Development

As more jobs become available and disappear at an alarming rate, a Learning Management System supports lifelong learning. It offers learners of all ages convenient and cheap opportunities to access content, certifications and trainings.

5.3.4 Reducing Environmental Impact Through Sustainable Practices

The incorporation of an LMS allows for the automation of many aspects of learning and less paper, books, and transportation to other learnings institutions. This greatly reduces carbon emissions that would ordinarily be expected of traditional learning methods.

5.3.4 Bridging Urban-Rural Divide

Through the help of its built-in AI technology and digital resources, the platform helps learners in remote or underdeveloped areas receive the “same” quality education as learners in developed urban areas.

6 APPENDIX

7 PUBLICATION

Our research paper titled **"Learning Management System using LLAMA"** got published in *Hinweis International Conference on Recent Trends in Engineering and Technology, RTET-2024*

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Date of Conference: 28th December 2024



Figure 7.1: Certificate of Paper Presentation and Publication

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